

TrES-3b

R-0372

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230628 · created 2026-07-18T17:12:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460123.8322 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.02
Transit depth (Rp/Rs)^2	1.69 +/- 0.53 [%]
Orbital Inclination (inc)	84.16 +/- 1.21
Airmass coefficient 1 (a1)	1.066 +/- 0.014
Airmass coefficient 2 (a2)	-0.057 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#2 - [531, 399]
Optimal Method	PSF photometry
Transit Duration (day)	0.0674 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.13 ± 0.02 vs 0.16309 (deviation 1.65σ)
Duration fit vs published	0.0674 d vs 0.0567 d (deviation 1.49σ)
Mid-time O–C	-3.4 min
Depth SNR	6.5
Residual scatter	1.36 %

Light curve

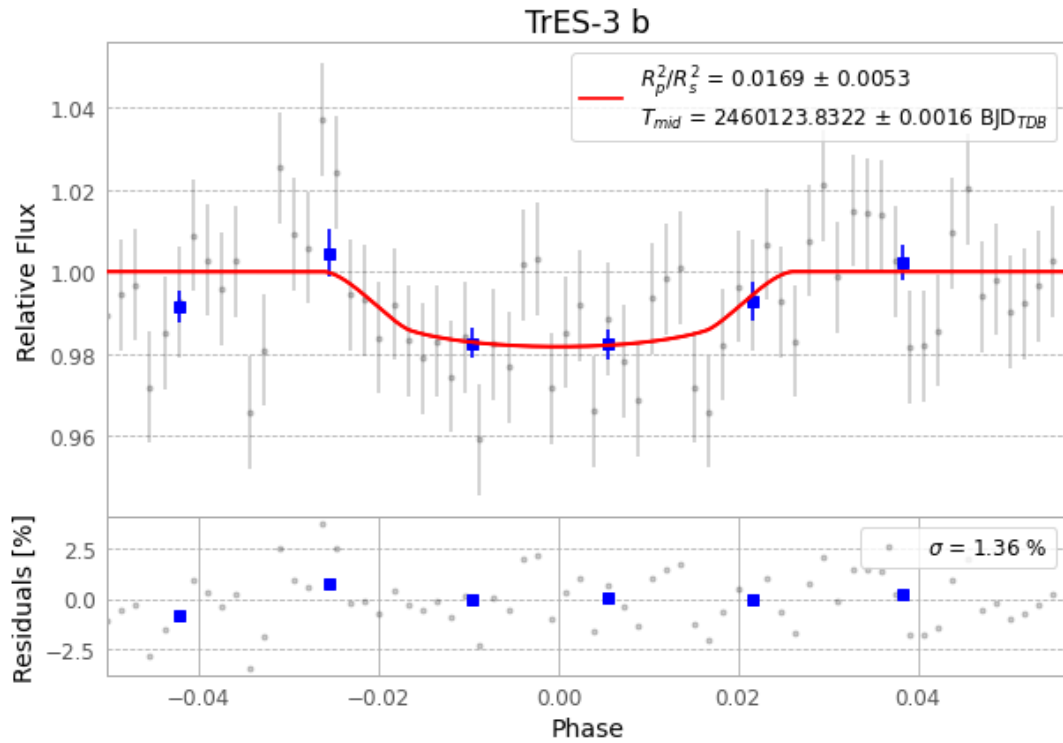


Figure 1 — FinalLightCurve_TrES-3 b_2023-06-27.png (SHA-256 f37e36512fb4d533...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d8cae6804ba6520e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit c09d8bf

Data provenance

68 FITS frames (45 MB), TRES-3230628061815.FITS → TRES-3230628093915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230628.log (66edba76af7b...), FinalLightCurve_TrES-3 b_2023-06-27.png (f37e36512fb4...), FinalLightCurve_TrES-3 b_2023-06-27.csv (4a9914341e5e...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 17:12Z.